

CS & IT ENGINEERING



Operating System

CPU Scheduling

Lecture - 3

By- Vishvadeep Gothi sir



Recap of Previous Lecture



Topic

SJF Scheduling

Topic

SRTF Scheduling

Topics to be Covered



Topic

LJF Scheduling

Topic

LRTF Scheduling

Topic

HRRN

Topic

Priority Based Scheduling



Topic : SRTF (Shortest Remaining Time First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	9	22	22	13
P2	1	6			
P3	2	4			
P4	3	2			
P5	6	1			

Response Time
0

	P1	P2	P3	P4	P3	P5	P3	P2	P1
0	1	2	3	5	6	7	9	14	22

[NAT]

#Q. Response time of processes in non-preemptive scheduling algorithms are equal to waiting time of processes?

✓ True or False

Justify your answer with appropriate explanation.

In preemptive system for a process P_i

Response time of $P_i \leq$ WT of P_i



Topic : LJF (Longest Job First)

→ Convoy effect

Scheduling Criteria: Highest BT process first | Tie breaker $\Rightarrow$ FCFS

Type of Algorithm: Non-preemptive



Topic : LJF (Longest Job First)

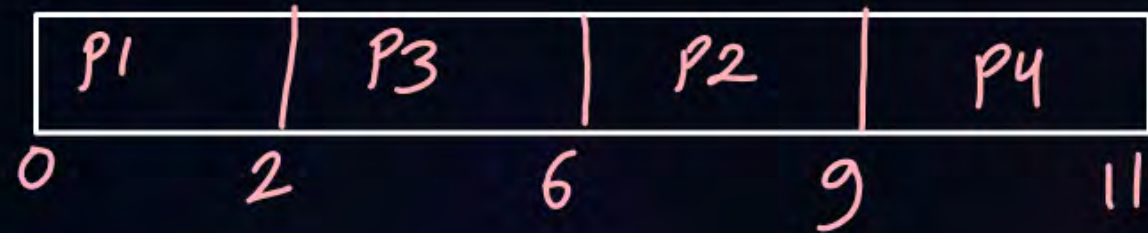
Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	6			
P2	0	3			
P3	0	4			
P4	0	2			





Topic : LJF (Longest Job First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	2			
P2	1	3			
P3	2	4			
P4	3	2			



LRTF (Longest remaining time First) :-

Preemptive LJF

scheduling criteria $\Rightarrow$ highest BT first | Tie breaker = FCFS

Type $\Rightarrow$ preemptive

ex:-

LRTF

	AT	BT
P1	0	3
P2	0	5
P3	0	1

} total
9

	BT
P1	3 2 1
P2	5 3 2 1
P3	1

P2	P1	P2	P1	P2	P1	P2	P3	
0	2	3	4	5	6	7	8	9

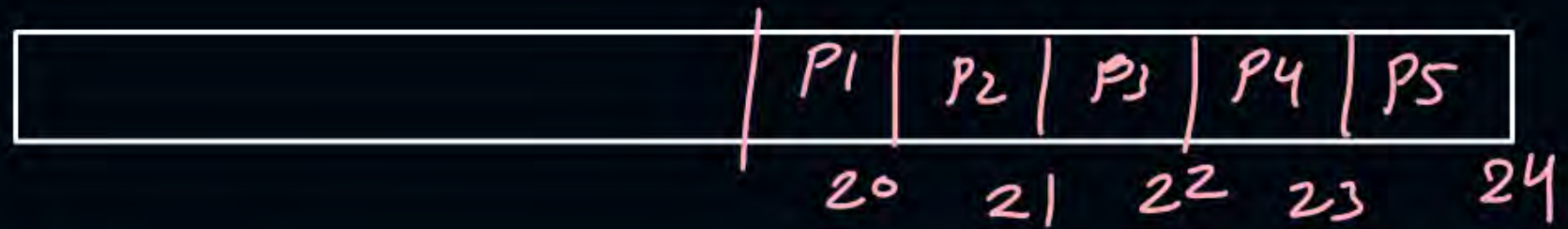
	P1	P2	P3
	7	8	9

LRTF :-

ex :-

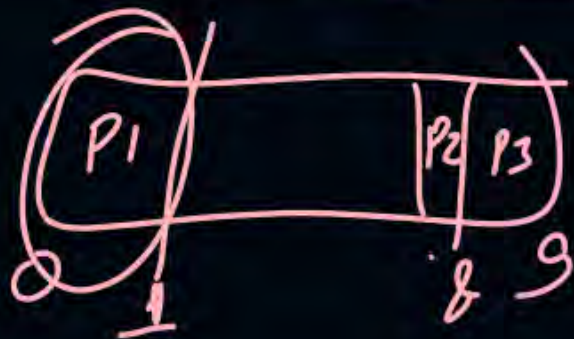
	AT	BT
P ₁	0	5
P ₂	0	3
P ₃	0	8
P ₄	0	2
P ₅	0	6

} Total 24



P ₁	0	1
P ₂	1	5
P ₃	2	3

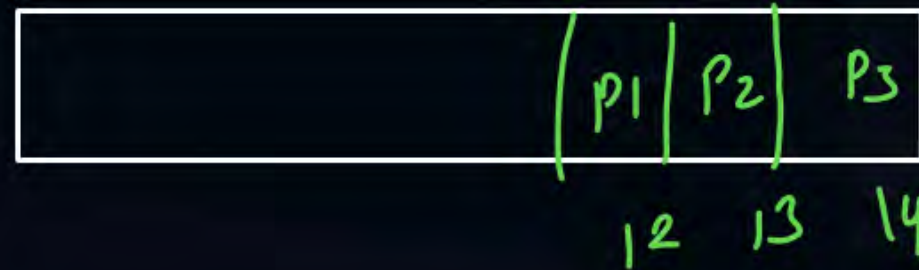
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#Q. Consider three processes (process id 0, 1, 2 respectively) with compute time bursts 2, 4 and 8 time units. All processes arrive at time zero. Consider the longest remaining time first (LRTF) scheduling algorithm. In LRTF ties are broken by giving priority to the process with the lowest process id. The average turn around time is:

	AT	BT	CT	TAT
P ₀	0	2	12	12
P ₁	0	4	13	13
P ₂	0	8	14	14

avg TAT = 13



A 13 units

B 14 units

C 15 units

D 16 units



Topic : HRRN (Highest Response Ratio Next)



Objective: Not only favors short jobs but decreases the WT of longer jobs.



Topic : HRRN (Highest Response Ratio Next)

Scheduling Criteria: ^{Highest} response ratio / Tie breaker $\Rightarrow$ SJF

Type of Algorithm: Non-preemptive

$$\text{Response Ratio} = \frac{W + S}{S}$$

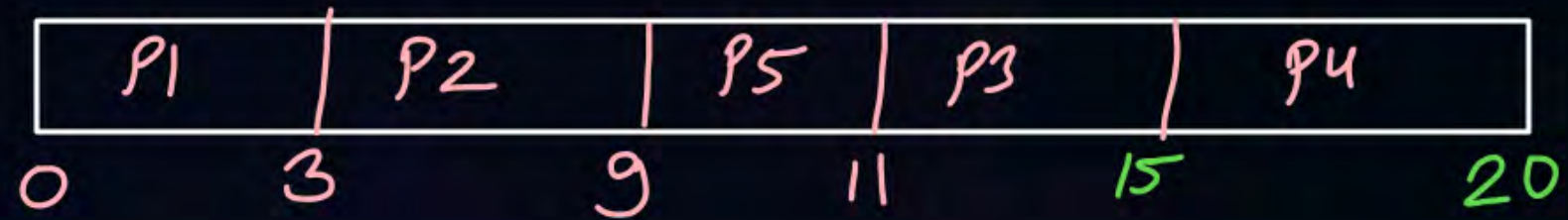
W = Wait Time of process till now
 S = Service/Burst Time



Topic : HRRN (Highest Response Ratio Next)

Process	Arrival Time	Burst Time
P1	0	3
P2	2	6
P3	4	4
P4	6	5
P5	8	2

STF:-

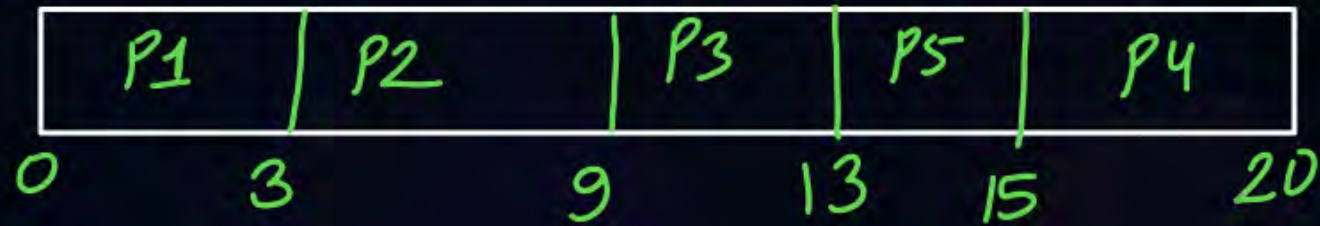




Topic : HRRN (Highest Response Ratio Next)

Process	Arrival Time	Burst Time
P1	0	3
P2	2	6
P3	4	4
P4	6	5
P5	8	2

HRRN :-



AT time 9 :-

$$RR(P3) = \frac{5+4}{4} = 2.25 \text{ (Highest)}$$

$$RR(P4) = \frac{3+5}{5} = 1.6$$

$$RR(P5) = \frac{1+2}{2} = 1.5$$

At time 13 :-

$$RR(P4) = \frac{7+5}{5} = 2.4$$

$$RR(P5) = \frac{5+2}{2} = 3.5 \text{ (Highest)}$$



Topic : Priority Based Algorithm

Scheduling Criteria: highest priority process first | Tie breaker $\Rightarrow$ given in Questⁿ

Type of Algorithm: Non-preemptive,
preemptive

Priority $\begin{cases} \rightarrow \text{static} & \text{fixed priority} \Rightarrow \text{our syllabus} \\ \rightarrow \text{dynamic} & \text{variable ''} \end{cases}$



Topic : Priority Based Algorithm

Process	Arrival Time	Burst Time	Priority
P1	0	4	4
P2	1	2	5
P3	2	3	6
P4	3	1	10(Highest)
P5	4	2	9
P6	5	6	7

Non-preemptive :-

P1	P4	P5	P6	P3	P2	
0	4	5	7	13	16	18



Topic : Priority Based Algorithm

Process	Arrival Time	Burst Time	Priority
P1	0	4	4
P2	1	2	5
P3	2	3	6
P4	3	1	10(Highest)
P5	4	2	9
P6	5	6	7

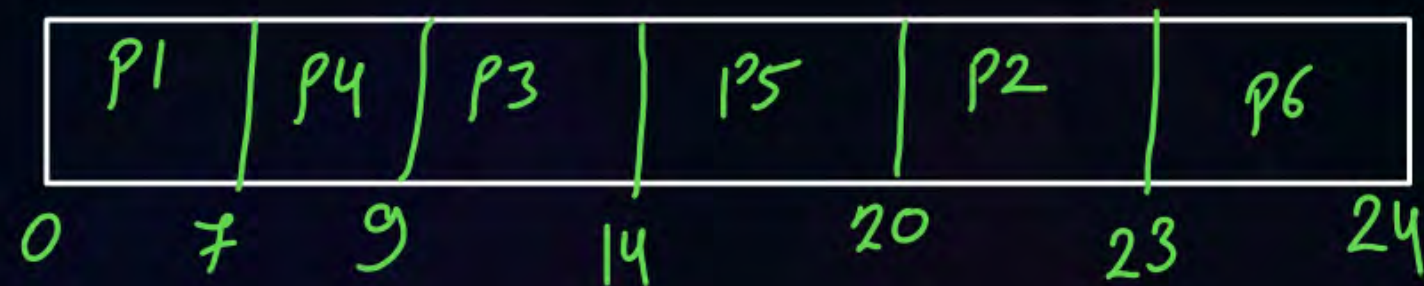
Preemptive :-

P1	P2	P3	P4	P5	P6	P3	P2	P1	
0	1	2	3	4	6	12	14	15	18



Topic : Priority Based Algorithm Question Non-Preemptive

Process	Arrival Time	Burst Time	Priority
P1	0	7	9
P2	1	3	4
P3	2	5	2
P4	3	2	1 (Highest)
P5	4	6	3
P6	5	1	8





Topic : Priority Based Algorithm Question Preemptive

Process	Arrival Time	Burst Time	Priority
P1	0	7	9
P2	1	3	4
P3	2	5	2
P4	3	2	1 (Highest)
P5	4	6	3
P6	5	1	8

p1	p2	p3	p4	p3	p5	p2	p6	p1	
0	1	2	3	5	9	15	17	18	24



Topic : Priority Based Algorithm

Advantages:

1. Better response for real time situations

Disadvantages:

2. Low Priority Processes may suffer from starvation

Solⁿ of starvation in priority algo:-

Aging :- (only possible with dynamic priority)

⇒ If a process waits for certain duration then its priority is increased by 1 periodically, hence at one time it will become highest priority process.



2 mins Summary

Topic

LJF Scheduling

Topic

LRTF Scheduling

Topic

HRRN

Topic

Priority Based Scheduling



Happy Learning

THANK - YOU